

1 Introduction

Multi-principal element alloys (MPEAs) represent a relatively new and innovative class of materials comprising three or more elements and showing remarkable mechanical properties superior to their pure or dilute counterparts, such as mechanical strength, fracture toughness, ductility, and resistance to hydrogen embrittlement. [1, 2, 3, 4]. The BCC refractory MPEAs of the family Cr-Mo-W-Nb-V-Ta-Ti-Zr-Hf have recently gained particular interest due to their exceptional high-temperature strength retention, exceeding the capabilities of current superalloys. [5, 6, 7, 8, 9, 10, 11, 12] With millions of possible compositions within just one single crystal, these complex systems offer significant potential to achieve tailored properties for specialized applications. However, navigating the immense multicomponent HEA compositional space to guide alloy discovery with optimized or desired properties poses significant challenges. Developing mechanistic multi-scale theories that connect atomistic-level phenomenon to microscopic material properties, such as temperature dependent yield stress, have been instrumental in exploring this design space. [13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23]

While theoretical models hold significant potential for predicting the macroscopic properties of MPEAs and navigating their vast design space, a key limitation lies in the computational cost of obtaining the necessary input parameters, which typically rely on atomistic simulations. This challenge is particularly acute for body-centered cubic (BCC) materials, where plasticity is controlled by the motion of screw dislocations [24, 21]. In contrast to edge dislocation theories, which offer a simplified approach by using misfit volume as a primary parameter [13, 14, 25], no such simplification exists for screw dislocation theories. As a result, these models require the computation of critical parameters via expensive atomistic simulations. Two key quantities in screw dislocation theories are the Peierls barrier and the solute-screw interaction energy [24, 21]. The Peierls barrier represents the intrinsic lattice resistance to dislocation motion, while the solute-screw interaction energy describes the influence of solute atoms on the dislocation’s behavior. Both factors serve as energy barriers to dislocation motion, and their calculation is further complicated by random solute fluctuations around the dislocation. Atomistic simulations are commonly used to determine these parameters in alloys [26], but applying them to multi-component alloys presents two significant challenges: (a) the vast design space of these alloys and (b) the need for numerous realizations to obtain a statistically accurate average due to the random solute environments. To address these limitations, machine learning (ML) and deep learning (DL) models offer a promising alternative by streamlining the computation of these input parameters, potentially reducing the reliance on computationally expensive brute-force methods.

The advent of ML and DL methods have revolutionized the ways of materials design, physical modeling, and properties measurement [27, 28, 29, 30, 31, 32]. These methods can discover hidden patterns in the training data and thus have been merged into all disciplines. The use of DL methods within the field of crystalline materials is wide, from machine learning interatomic potential development [33, 34, 35, 36, 37] to intrinsic properties calculation [38] to dynamic crack path prediction [39]. Among all the DL architectures, GNN models have been developed to deal with graph structures which model a series of objects (nodes) and the relationships between them (edges), making them an ideal solution for crystalline materials where the atoms are nodes and the edges represent the intermediate bond [40, 41].

While machine learning (ML) methods have accelerated the exploration of multi-principal element alloys (MPEAs), they often target specific material properties in isolation. This narrow focus can limit their ability to incorporate the broader, interdisciplinary knowledge that is critical for true breakthroughs in alloy design and scientific discovery. To overcome these limitations, multi-agent systems have emerged as a transformative approach, facilitating the integration of multi-modal data and external knowledge—such as new developments in physics and materials science including theoretical models—into the design process in a more holistic and adaptive way [42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52]. In multi-agent systems, a group of AI agents—each powered by Large Language Models (LLMs)—work together dynamically to solve complex, multi-faceted problems. By assigning distinct roles to each agent via targeted prompts, the system can divide and conquer different aspects of the alloy design challenge.

In this paper, we present an LLM-based multi-agent system that leverages the power of: (a) a suite of advanced Large Language Models (LLMs), responsible for fundamental tasks such as planning, reasoning, and decision-making; (b) a group of specialized agents, each playing distinct roles within the system; and (c) a set of external tools for various tasks, including a newly developed Graph Neural Network (GNN) model to predict atomic-level material properties. This multi-agent model builds upon AtomAgents, a multimodal multi-agent system with advanced simulation capabilities for alloy design and discovery. AtomAgents was capable of retrieving new physics from atomistic simulations, solving complex multi-faceted alloy discovery problems. However, as the complexity of the alloy systems increased, the computational cost of atomistic simulations became prohibitive.